

Project Proposal

1 Project Title

- Stock Signal Classification Using Technical Indicators and News Sentiment (Buy/Hold/Sell)

2 Team Members

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3 Project Details

3.1 Project Objective

- **What is the project Objective?**

The objective is to build a machine learning system that predicts a stock's next-day price movement signal (Buy/Hold/Sell) using historical OHLCV market data (Open/High/Low/Close/Volume), engineered technical indicators, and aggregated daily news sentiment. The model will be evaluated using time-based splits and a simple backtest to compare performance with baseline approaches.

- **What problem to solve?**

Stock market decisions are difficult because prices are noisy, and emotional bias can lead to inconsistent decisions. We want an automated approach that produces a consistent signal:

- Buy if the model expects a sufficiently positive next-day return
- Sell if the model expects a sufficiently negative next-day return
- Hold if the next-day return is likely small or uncertain

- **Why the problem important?**

This problem represents a real-world machine learning scenario involving:

- time-series prediction under noisy conditions
- avoiding data leakage (future information accidentally used in training)
- realistic evaluation (time-based testing and backtesting rather than only accuracy)
- practical model comparison (ML vs. simple technical rule baselines)

- **Why can machine learning help to solve the problem?**

Hand-crafted trading rules (example: $RSI < 30 \rightarrow$ buy) can be brittle and fail across different market regimes. Machine learning can:

- learn non-linear interactions among indicators
- incorporate news sentiment as an additional signal
- be tested objectively against baselines like buy-and-hold and moving-average strategies

(This project is for educational research and is not financial advice.)

3.2 Datasets

We will use two datasets merged by date (and ticker for stock prices).

- **What is the data and where you obtain it?**

A) Stock Price Dataset (OHLCV)

Daily OHLCV price data for 20–30 large-cap U.S. stocks (examples: AAPL, MSFT, AMZN, GOOG, META, NVDA, TSLA, JPM, XOM, etc.).

- Source: Yahoo Finance download (via yfinance or CSV export)

B) News Headlines Dataset (Sentiment)

A Kaggle dataset containing dated news headlines (e.g., datasets used for “Daily News for Stock Market Prediction” / “financial news headlines with date”).

- Source: Kaggle download as CSV (headlines + date/time)

- **How is the data collected?**

A) Stock Price Dataset (OHLCV)

For each ticker, we download daily OHLCV data over approximately 2014–2026 (or the largest common time window available). Data is cleaned to remove missing days and aligned by market dates.

B) News Headlines Dataset (Sentiment)

Headlines are grouped by day, sentiment is computed per headline, then aggregated per day. We use lagged sentiment (previous day) to avoid leakage into next-day prediction.

- **What will be the features and labels you will use?**

Features (X)

A) Technical Indicator Features (from OHLCV):

- Returns: 1-day, 5-day, 10-day returns
- Moving averages: SMA/EMA (5, 10, 20, 50)
- MA crossover flags (e.g., $SMA_{10} > SMA_{20}$)
- Volatility: rolling standard deviation of returns (5, 10, 20 days)

- RSI(14)
- MACD and signal line values
- Bollinger Band features (upper/lower bands, %B)
- Volume features: volume change %, rolling volume averages

B) News Sentiment Features (from headlines):

We compute sentiment per headline (FinBERT preferred; fallback: VADER/TextBlob if needed) and aggregate daily:

- mean sentiment score (daily)
- sentiment standard deviation (daily)
- number of positive/negative/neutral headlines per day
- total headline count per day (news volume)

Leakage prevention:

We will only use sentiment information known *before* the target day.

- For predicting day $t+1$, we use sentiment from day t (lagged by 1 day).

Labels (y)

We use a **3-class Buy/Hold/Sell label** based on next-day return:

Let return be:

$$r_{t+1} = \frac{Close_{t+1} - Close_t}{Close_t}$$

- Buy (class +1): $r_{t+1} > +0.003$ (greater than +0.3%)
- Sell (class -1): $r_{t+1} < -0.003$ (less than -0.3%)
- Hold (class 0): otherwise

This threshold reduces noisy “tiny move” days being treated as meaningful.

• How many examples for training, validation, and testing?

Because this is time-series, we do **time-based splits (no shuffling)**:

- **Train:** first 70% of dates
- **Validation:** next 15%
- **Test:** last 15% (most recent period)

Optional enhancement (if time permits): walk-forward evaluation (rolling training window).

Number of Examples (Expected)

Using ~ 10 years of daily data:

- 1 ticker $\approx$ 2,500 rows
- With **25 tickers**, total $\approx 25 \times 2,500 = 62,500$ rows

After removing indicator warm-up rows and missing values, we expect roughly **55,000–60,000 usable samples**.

Approximate split:

- Train: $\sim 40k$
- Validation: $\sim 8-9k$
- Test: $\sim 8-9k$

3.3 Machine Learning Algorithm

Main Model

Gradient Boosted Decision Trees (XGBoost or LightGBM)

Why this algorithm?

- Strong performance on tabular data (engineered indicators + sentiment aggregates)
- Captures nonlinear feature interactions
- Efficient training and tuning
- Often outperforms deep models on structured feature sets with limited compute

Baselines (required comparisons)

1. **No-skill baseline:** always predict “Hold” or majority class
2. **Logistic regression / softmax regression:** interpretable baseline
3. **Rule-based baseline:** e.g., moving average crossover or RSI rule
4. **Buy-and-hold benchmark:** used for backtest comparison

Key Experiment (Ablation Study)

We will train and compare:

- **Model A:** technical features only
- **Model B:** technical + sentiment features

This directly measures whether sentiment adds predictive value.

Evaluation Metrics

Classification metrics:

- Accuracy
- Macro F1-score (important for 3-class imbalance)
- Confusion matrix

Backtesting metrics (simple signal strategy):

- cumulative return vs buy-and-hold
- max drawdown
- number of trades
- optional transaction cost assumption (e.g., 0.05% per trade)

3.4 Expected Outcomes

Expected outcomes

- A complete pipeline to:
 - download OHLCV data for multiple tickers
 - compute technical indicators
 - compute daily news sentiment aggregates
 - merge features by date and apply lagging to prevent leakage
 - train and evaluate classification models
- A trained model that outputs a Buy/Hold/Sell signal for next-day movement.
- Results comparing:
 - boosted trees vs baselines
 - technical-only vs technical+sentiment

Success criteria

- Model achieves better Macro F1 (and/or accuracy) than baselines on the test period.
- Backtesting shows a clear, measurable comparison vs simple strategies and buy-and-hold.
- The final report includes conclusions about when sentiment helps (e.g., high headline volume days) and when it does not.